



# Farmland Investment and Portfolio Choice: A Replication and Extension of Lohano & King (2009)

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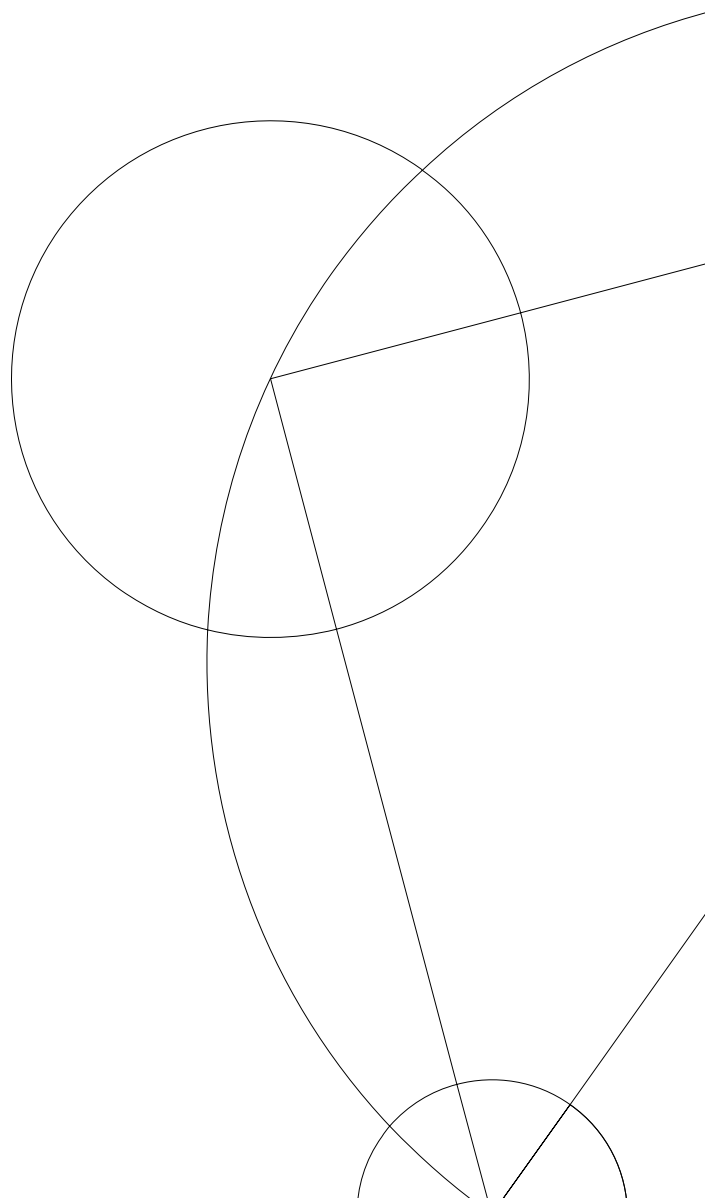
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## **Abstract**

This paper replicates and extends the stochastic dynamic programming model of farmland investment and portfolio choice by Lohano and King (2009). The model studies how farmers optimally allocate wealth between farmland, financial assets, borrowing, and risk-free savings under uncertainty regarding farming returns, farmland prices, and financial market returns. We implement the original model in Python and reproduce the main policy functions and simulation results from the original study, including optimal farmland investment decisions, wealth accumulation, and probabilities of exiting farming.

Using the replicated framework, we conduct counterfactual analyses to examine the roles of borrowing constraints, transaction costs, and economies of scale. The results show that relaxing borrowing constraints mainly increases the scale and riskiness of investment decisions, while transaction costs generate inaction in farmland adjustment. When transaction costs are removed, land trading becomes more active and long-run farmland holdings increase. Furthermore, we extend the model by introducing economies of scale in production costs, which significantly increases optimal farm size, expected wealth, and farm survival rates, while also increasing wealth dispersion across farmers.

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# 1 Introduction

Being a farmer involves making several important investment decisions. The farmer must decide how much land to operate, how much debt to take on, and how to allocate wealth between farm and nonfarm assets. These decisions are not simply made one period at a time, since the choices made today affect both current wealth and future investment opportunities. Farmland investment can therefore be viewed as a dynamic decision problem.

This paper follows Lohano and King (2009), who model farmland investment and financial management as a stochastic dynamic programming problem. In their model, a farmer optimally chooses how to invest in farmland and financial assets over a 20-year planning horizon. Specifically, the farmer chooses how much farmland to buy or sell, how much to invest in a risky mutual fund, and whether remaining funds should be borrowed or invested in a risk-free asset. The model also includes transaction costs and borrowing constraints, which are important because farmland is costly to adjust and is often financed through debt. The model is therefore useful because it captures how current investment decisions depend on future uncertainty, financial constraints, and the value of waiting.

Our main motivation is to understand how uncertainty, credit constraints, and transaction costs affect optimal farmland investment, wealth accumulation, and the probability of exiting farming. A further motivation for the project is computational. The original model is implemented in MATLAB and consists of several files, which makes the code difficult to follow and modify. As part of the replication, we therefore implement the model in Python with the aim of making the code more manageable and easier to extend. This also makes it easier to conduct counterfactual simulations and to understand how the different parts of the dynamic programming problem interact.

In this paper, we first replicate the baseline model of Lohano and King (2009) and then use our Python implementation to conduct counterfactual simulations. In particular, we examine how the results change when borrowing constraints are tightened and when transaction costs are removed, and examine the effects of adding economies of scale to the model.

The remainder of this paper is structured as follows. Section 2 presents the theoretical model and the Bellman equation underlying the farmer's optimization problem. Section 3 describes how the model is solved numerically, as well as the computational challenges we faced during implementation. Section 4 presents the replication results and compares our findings to Lohano and King (2009). Section 5 introduces our counterfactual results. Finally, Section 6 concludes and discusses the broader implications and limitations of the model.

## 2 The Model

### 2.1 Economic setup and variables

The model follows Lohano and King (2009), who formulate a stochastic dynamic optimization problem where a farmer makes investment and portfolio decisions over time. The farmer's objective is to maximize expected utility of terminal wealth  $W_{T+1}$ :

$$\max_{\{(x_t, x_{mt})\}_{t=0}^T} E_0[u(W_{T+1})] \quad (1)$$

We assume a CRRA utility function, which is formally defined in Section 2.4. The investment decisions are affected by several factors including returns from farming, changes in farmland prices, returns from financial assets, borrowing opportunities and transaction costs.

The state of the system at time  $t$  is given by four variables: the return from farming  $R_t$ , the farmland price  $P_t$ , the amount of land owned  $L_t$ , and the farmer's wealth  $W_t$ . Together, these variables form the state vector  $(R_t, P_t, L_t, W_t)$  and describe the farmer's current situation, which affects the decisions made in each period (see Table 9 in Appendix A.1 for a full list of variables and parameters).

In each period, the farmer chooses two control variables,  $x_t$  and  $x_{mt}$ . The variable  $x_t$  denotes how much land the farmer decides to buy or sell, and  $x_{mt}$  denotes how the farmer allocates his wealth between a risky financial asset and a risk-free asset. The choices must lie in feasible sets denoted by  $X_t$  and  $X_{mt}$ .

### 2.2 Stochastic processes and wealth dynamics

The farmer's decisions are made under uncertainty about returns from farming, farmland prices, and financial assets. These factors affect the investment decisions and evolve stochastically over time.

The three variables  $R_t$ ,  $P_t$  and  $M_t$  follow different stochastic processes. Farming returns  $R_t$  follow a standard AR(1) process, farmland prices  $P_t$  follow an autoregressive process that also depends on past farming returns, and mutual fund returns  $M_t$  are assumed to be i.i.d. around a constant mean. The terms  $\epsilon_{1,t+1}$ ,  $\epsilon_{2,t+1}$ , and  $\epsilon_{3,t+1}$  denote random shocks that capture unexpected changes in the three variables. The shocks are assumed to be normally distributed with mean zero. The variables are expressed in logarithms, which ensures positive values and allows the equations to capture proportional changes

over time. The stochastic processes are given by:

$$\ln R_{t+1} = \beta_0 + \beta_1 \ln R_t + \epsilon_{1,t+1} \quad (2)$$

$$\ln P_{t+1} = \alpha_0 + \alpha_1 \ln P_t + \alpha_2 \ln R_t + \epsilon_{2,t+1} \quad (3)$$

$$\ln M_{t+1} = \gamma_0 + \epsilon_{3,t+1} \quad (4)$$

Equation (2) describes farming returns, which depend on past returns and a random shock. This means that high returns today are likely to be followed by higher returns next period when  $\beta_1 > 0$ . Equation (3) describes farmland prices, which depend on past prices, past farming returns, and a random shock. This means that farmland prices are affected by both earlier prices and by how profitable farming is. Equation (4) describes mutual fund returns, which vary around a constant mean and are affected by random shocks. This means that financial returns are uncertain and do not depend on past values. Since  $M_t$  is assumed to be i.i.d., it does not contain information about future values and therefore does not enter the state vector.

Together with the stochastic processes above, the farmer's choices determine the transition equations for land holdings  $L_{t+1}$  and wealth  $W_{t+1}$ . The transition equation for land holdings is given by:

$$L_{t+1} = L_t + x_t \quad (5)$$

Here,  $x_t$  denotes net land purchases. If  $x_t > 0$ , the farmer buys land, while if  $x_t < 0$ , the farmer sells land.

The farmer's cash balance is defined as total wealth minus the value of land holdings:

$$A_t = W_t - (P_t + \kappa - tc_s)L_t \quad (6)$$

The cash balance can be positive or negative. When  $A_t < 0$ , the farmer is in a debt position (borrowing), while  $A_t > 0$  indicates that the farmer has savings. The transition equation for wealth is then given by:

$$\begin{aligned} W_{t+1} = & (1+r)[A_t - (P_t + \kappa + tc)x_t - cL_{t+1} - x_{mt}] \\ & + M_{t+1}x_{mt} + R_{t+1}L_{t+1} + (P_{t+1} + \kappa - tc_s)L_{t+1} \end{aligned} \quad (7)$$

The interest rate depends on whether the farmer borrows or lends:

$$r = \begin{cases} r_b, & \text{if } A_t - (P_t + \kappa + tc)x_t - cL_{t+1} - x_{mt} < 0, \\ r_l, & \text{otherwise.} \end{cases} \quad (8)$$

Transaction costs depend on whether land is purchased or sold:

$$tc = \begin{cases} tc_p, & \text{if } x_t > 0, \\ -tc_s, & \text{otherwise.} \end{cases} \quad (9)$$

Equation (7) describes the law of motion for the farmer's wealth. The equation can be divided into two parts: the use of current cash balance and the sources of next-period wealth.

The farmer's cash balance  $A_t$  is used to finance land purchases  $(P_t + \kappa + tc)x_t$ , production costs  $cL_{t+1}$ , and financial investments  $x_{mt}$ . After these expenditures, the remaining amount represents the net cash position carried into the next period. This amount is multiplied by  $(1 + r)$ , reflecting the return on these funds. As described in equation (8), if this amount is negative, the farmer must borrow at interest rate  $r_b$ . If it is positive, the farmer earns the lending rate  $r_l$ . Since  $r_l < r_b$ , the farmer faces higher costs when in a debt position.

The final part of equation (7) describes the sources of next-period wealth. This includes the risky financial asset  $M_{t+1}x_{mt}$ , income from farming  $R_{t+1}L_{t+1}$ , and the value of land holdings  $(P_{t+1} + \kappa - tc_s)L_{t+1}$ . Therefore, the farmer's wealth in the next period is increased by financial returns, farming income, and future land value. As described in equation (9), transaction costs depend on whether the farmer buys or sells land. When the farmer purchases land he pays  $tc_p$  and when he sells land he pays  $tc_s$ . This affects both current land purchases and the future value of land holdings.

## 2.3 Constraints

The farmer's decisions are subject to several constraints that limit the choices available in each period. These constraints reflect the current land holdings, borrowing opportunities, and the available financial resources.

Firstly, land holdings must remain non-negative, implying that the farmer cannot sell more land than currently owned. Since land evolves according to  $L_{t+1} = L_t + x_t$ , as described in Equation (5), the constraint can be written as:

$$x_t \geq -L_t$$

Furthermore, investment in the mutual fund must be non-negative. The economic intuition behind this is that if the farmer could invest a negative amount in the mutual fund, he would essentially be shorting the asset. Since this is not allowed, investment must be

non-negative:

$$x_{mt} \geq 0$$

Borrowing is limited by a debt-to-farm-asset (DFA) ratio, which is defined as:

$$DFA_t = \max \left\{ 0, \frac{-A_t}{(P_t + \kappa - tc_s)L_t} \right\}. \quad (10)$$

where  $A_t$  is the farmer's cash balance, defined in Equation (6). When  $A_t \geq 0$ , the DFA ratio is zero because the farmer has no debt. When  $A_t < 0$ , the farmer is in a debt position, and the ratio measures debt relative to the value of farm assets. In this case, borrowing is limited by the constraint  $DFA_t \leq \rho$ , where  $\rho$  denotes the maximum allowed DFA ratio at the time of the farmland purchase. This constraint captures that the farmer cannot finance land purchases with unlimited debt. However, they are allowed to have a  $DFA_t > \rho$ , but is only allowed in the sense that farmland assets decrease in value after the purchase, making  $DFA_t > \rho$ . In the baseline model, Lohano and King (2009) set  $\rho = 0.70$ , which implies that the farmer can borrow up to 70% of the value of farm assets. In our counterfactual section, we examine how changing  $\rho$  affects the farmer's investment decisions and model outcomes.

The farmer can either voluntarily exit farming by selling all land holdings, or be forced to sell all land if wealth becomes negative. In both cases, all land is liquidated:

$$x_t = -L_t$$

If  $W_t < 0$ , this represents bankruptcy. If  $W_t \geq 0$ , it represents voluntary exit. After exiting farming, re-entry is not allowed for the remaining planning horizon. This implies that once all land holdings have been sold, future land holdings must remain zero:

$$L_{t+s} = 0, \quad s = 1, 2, \dots, T - t$$

Lastly, the farmer's choices must satisfy  $x_t \in X_t$  and  $x_{mt} \in X_{mt}$ , where  $X_t$  and  $X_{mt}$  denote the feasible choice sets defined by the constraints above.

## 2.4 Utility function and Bellman equation

The farmer's objective, stated in Equation (1), is to choose land and portfolio decisions in each period in order to maximize expected utility of terminal wealth. This utility is

described by a CRRA (power) utility function:

$$U(W) = \frac{W^{1-\theta} - 1}{1-\theta}, \quad \theta \geq 0 \quad (11)$$

where  $\theta$  is the coefficient of relative risk aversion. When  $\theta = 0$ , the farmer is risk neutral, while  $\theta > 0$  implies that the farmer has risk-averse preferences. Thus, a higher value of  $\theta$  implies that the farmer is more risk averse. Lohano and King (2009) solve the model for both the risk-neutral case ( $\theta = 0$ ) and a risk-averse case ( $\theta = 1$ ). In our replication, we follow the same approach and present results for both cases.

However, the CRRA utility function is not defined for negative wealth. In order to allow for all possible values of  $W$ , the utility function is specified as:

$$u(W) = \begin{cases} W, & \text{if } \theta = 0, \\ U(W), & \text{if } \theta > 0 \text{ and } W \geq b, \\ \frac{U(b)}{b}W, & \text{if } \theta > 0 \text{ and } W < b, \end{cases} \quad (12)$$

where  $b > 0$  is a threshold level of wealth. For wealth below this threshold, utility is linear (risk-neutral) while for higher wealth levels ( $W \geq b$ ) it follows the standard CRRA form (risk-averse). This allows the model to handle low or negative wealth states, including the bankruptcy case. In the baseline model, Lohano and King (2009) set  $b = 60,000$ .

The dynamic optimization problem can be written recursively using the Bellman equation:

$$V_t(R_t, P_t, L_t, W_t) = \max_{(x_t, x_{mt})} E[V_{t+1}(R_{t+1}, P_{t+1}, L_{t+1}, W_{t+1}) | R_t, P_t, L_t, W_t] \quad (13)$$

The Bellman equation represents the farmer's dynamic optimization problem. Given the current state  $(R_t, P_t, L_t, W_t)$ , the farmer chooses how much land to buy or sell and how much to invest in the mutual fund ( $x_t$  and  $x_{mt}$ ). The farmer chooses these controls in order to maximize the expected continuation value, which depends on next period's state  $(R_{t+1}, P_{t+1}, L_{t+1}, W_{t+1})$ . The continuation value is determined by the stochastic processes given in Equations (2)–(4) and the transition equations described in Equations (5) and (7).

At the end of the planning horizon,  $T + 1$ , there are no future decisions left. Therefore, the value function equals utility from terminal wealth:

$$V_{T+1}(R, P, L, W) = u(W) \quad (14)$$

The problem is solved by backward induction. Starting from the terminal condition  $V_{T+1}$ , the value function is computed backwards period by period as  $V_T, V_{T-1}, \dots, V_0$ . At each step, the optimal policy is obtained by solving the maximization problem in the Bellman equation, which gives the optimal choices of  $x_t$  and  $x_{mt}$  for each state.

### 3 Solving the Model

In this section, we describe how the dynamic model is solved numerically. We closely follow the methodology of Lohano and King (2009).

The model presented in section 2 is a finite-horizon stochastic dynamic model with two joint continuous control variables. Solving the model involves solving the Bellman equation. We use backwards induction where the Bellman equation is solved from the terminal period to the initial period. At each time step, the value function is approximated numerically, while the optimal controls are obtained by solving the Bellman maximization problem over a discretized control space. Here the value function is defined recursively, with the terminal condition given by the utility of terminal wealth.

Since the model does not have a closed-form solution and includes continuous state variables, the value function needs to be approximated numerically. Following Lohano and King (2009), the value function is approximated using a collocation method with linear spline basis functions defined over a set of nodes in the state space. In particular, the value function is approximated as a linear combination of basis functions,

$$V_t(R, P, L, W) \approx \sum_{i=1}^N c_{it} \phi_i(R, P, L, W),$$

where  $\{\phi_i(\cdot)\}_{i=1}^N$  are basis functions and  $\{c_{it}\}_{i=1}^N$  are coefficients to be determined. The collocation method determines these coefficients by enforcing the Bellman equation at a specified set of collocation nodes. That is, the approximate value function is required to satisfy the Bellman equation exactly at these points. At each time step of the backward-induction procedure, the Bellman equation is evaluated at the collocation nodes, while interpolation is used to approximate the value function between nodes. The choice of nodes is therefore crucial, as it determines how accurately the value function is approximated over the remaining state space. This approach relies on the value function being sufficiently smooth, and is less affected by the curse of dimensionality than full grid-based methods. Following Lohano and King (2009), we use seven nodes for  $R_t$ , seven nodes for  $P_t$ , five nodes for  $L_t$  and 21 nodes for  $W_t$ , which are all uniformly space over the state space.

At each state, the continuous control variables are discretized into finite grids, and the value function is evaluated for all possible combinations. The optimal policy is then selected as the combination that maximizes the expected value. Hence, we discretize the control space  $X_t$  into 41 uniform levels and  $X_{mt}$  into 25 uniform levels. At each time step, the value function is approximated numerically, while the optimal controls are obtained by solving the Bellman maximization problem over a discretized control space, which is particularly important given the potential non-convexities introduced by constraints and

transaction costs. Lastly, the expectations in the Bellman equation are computed using Gauss-Hermite quadrature. For the model with mutual fund investment, 3 nodes are used for each of three stochastic error terms, and for the model without mutual fund investment, 5 nodes are used for each of the two stochastic error terms.

### 3.1 State space

In order to approximate the value function, the state space is restricted to the following ranges:  $230 \leq R_t \leq 540$ ;  $1,010 \leq P_t \leq 2,840$ ;  $400 \leq L_t \leq 2,000$ ; and  $0 \leq W_t \leq 6,000,000$ . These ranges are determined by Lohano and King (2009) and for  $R_t$  and  $P_t$  they are based on a 90% region for their distributions. The lower bound for wealth is the bankruptcy condition ( $W_t < 0$ ), and thus the condition for staying in farming is  $W_t \geq 0$ . For the parameter in the utility function we set  $b = 60,000$ .

The value function is defined over the state space  $400 \leq L_t \leq 2000$ , but must also be evaluated at  $L_{t+1} = 0$  since the farmer can exit by selling all farmland. Exit is irreversible, and without farmland no borrowing is possible. In this case, wealth is allocated to non-farm assets, implying  $V(W_{t+1}) = \mathbb{E}[u(W_{t+1})]$ . In the absence of mutual funds, all wealth is invested in the risk-free asset, so the value reduces to  $u(W_{T+1})$ .

Net wealth  $W_{t+1}$  may fall outside the state space due to shocks. If  $W_{t+1} < 0$ , bankruptcy occurs, the farm is liquidated ( $L_{t+1} = 0$ ), and wealth evolves at the borrowing rate. If  $W_{t+1}$  exceeds the upper bound, excess wealth is invested in non-farm assets (or the risk-free asset in the restricted model) and compounded forward. To ensure continuity,  $V(0) = u(0)$ .

To keep states within bounds, gross returns are truncated as  $R_{t+1} = \max\{\underline{R}, \min[g(R_t, \varepsilon_{1,t+1}), \bar{R}]\}$ , assigning probability mass outside the domain to the boundary. This ensures that simulated returns remain within the predefined state-space limits. In practice, if a simulated value falls below the lower bound ( $\underline{R}$ ), it is set equal to  $\underline{R}$ , while values above the upper bound ( $\bar{R}$ ) are set equal to  $\bar{R}$ . The same truncation is applied to farmland prices  $P_{t+1}$ .

## 3.2 Data and estimation

### 3.2.1 Estimation

In order to solve the model, equations (2)-(5) need to be estimated. Hence, the parameter of interest, as in Lohano & King (2009), is gross returns from crops per acre ( $R_t$ ), farmland price ( $P_t$ ), mutual fund returns ( $M_t$ ). The data used for estimating gross returns from crops per acre are published in Southwestern Minnesota Farm Business Association. The data used for estimating farmland prices are from Minnesota Rural Real Estate Market. For the period 1967-1992 these prices are found in Asheim and Brekke (1993) and for the

period 1990-2007 are from Taff (2008). Gross returns and farmland prices are adjusted for inflation to year 2000. For the mutual fund returns, inflation-adjusted compound annual total return for the S&P 500 in the period 1967-2007 have been used. We have been fortunate enough that the authors of the original paper have shared the data they used with us. The estimation results for equation (2)-(5) can be found in appendix A.2-4.

### 3.2.2 Parameters for $W_t$

Parameter values for the net wealth equation are the same as in Lohano and King (2009), where parameter are calibrated using SWMFBMA data and Nordquist (1983-2007). Machinery value is set to  $\kappa = 300$ , transaction costs are 1% for land purchases and 6% for selling land (machinery), and the maximum debt-to-farm-asset ratio is  $\rho = 0.7$ . The average cost of production for a 50-50 corn-soybean mix is \$201 per acre, with an additional \$30 opportunity cost, implying total costs of  $c = 231$ , while interest rates are set to 6% for borrowing and 3% for lending.

## 3.3 Implementation and challenges

When implementing the model from Lohano and King (2009) in Python, we faced a significant computational challenge: run time. In our first implementation, solving the full model took up to 3 hours. This led us to believe that the model was strongly affected by the curse of dimensionality, although this computational issue is not discussed in detail in the original paper.

|                                             | No mutual fund                                   | With mutual fund                                     |
|---------------------------------------------|--------------------------------------------------|------------------------------------------------------|
| Planning horizon                            | 20 years                                         | 20 years                                             |
| Number of state variables                   | 4                                                | 4                                                    |
| Number of control variables                 | 1                                                | 2                                                    |
| Collocation nodes for $R_t$                 | 7                                                | 7                                                    |
| Collocation nodes for $P_t$                 | 7                                                | 7                                                    |
| Collocation nodes for $L_t$                 | 5                                                | 5                                                    |
| Collocation nodes for $W_t$                 | 21                                               | 21                                                   |
| Total collocation nodes                     | 5,145                                            | 5,145                                                |
| Grid points for farmland choice $x_t$       | Stage 1: 41, stage 2: 21                         | 41                                                   |
| Grid points for mutual fund choice $x_{mt}$ | —                                                | 25                                                   |
| Control combinations per state              | $42 + 21 = 62$                                   | $41 \times 25 = 1,025$                               |
| Stochastic shocks integrated over           | 2                                                | 3                                                    |
| Quadrature nodes per shock                  | 5                                                | 3                                                    |
| Integration nodes per expectation           | 25                                               | 27                                                   |
| Approx. Bellman evaluations per period      | $5,145 \times 62 \times 25 \approx 7.97$ million | $5,145 \times 1,025 \times 27 \approx 142.4$ million |
| Approx. Bellman evaluations over 20 periods | $\approx 159.5$ million                          | $\approx 2.85$ billion                               |

**Table 1:** Computational setup for both models

In table 1, we have listed the computational setup for the original implementation and its resulting Bellman evaluations. Noticeably, the model with mutual fund has approximately 2.85 billion Bellman evaluations coming from approximately 159.5 million Bellman evaluations in the model without mutual fund. The increase is driven by introducing a new control variable and switching from a two-stage grid search method to a one-stage

grid search method.

To circumvent the curse of dimensionality, we therefore needed to make our initial implementation more efficient. First, we considered reducing the total number of Bellman evaluations by using a two stage grid search method in the model with mutual fund. This method first coarsely searches over the entire state space to identify a possible region for the global optimum. Then, at the promising area it defines a fine grid to find an optimum. Importantly, the effectiveness of a two-stage grid search depends on the quality of the coarse grid. If the initial grid is too sparse, it may fail to identify the region containing the global optimum, and the subsequent refinement will only improve a suboptimal candidate. When attempting this approach, we failed to find grid sizes that would replicate the results while also improving running times.

We then moved on to try vectorizing our initial implementation and compiling computationally intensive loops using the Numba library in Python. Vectorization allowed us to replace explicit Python loops with optimized array operations implemented in low-level compiled code through NumPy. This substantially reduced interpreter overhead associated with repeatedly evaluating the Bellman equation over large state and control grids. However, several parts of the model still relied on nested loops and conditional logic that were difficult to fully vectorize.

To address this, we used Numba’s just-in-time (JIT) compilation capabilities to compile sections of the code into machine code during runtime. In particular, the Bellman iteration and grid search procedures benefited from JIT compilation, as these routines involved millions of repeated numerical operations. Combining vectorization with Numba reduced total run time considerably while preserving the numerical accuracy of the model solution. This allowed us to solve the extended model with mutual funds within a feasible computational time frame despite the large increase in Bellman evaluations. In table 2 run times and speed-up can be found.

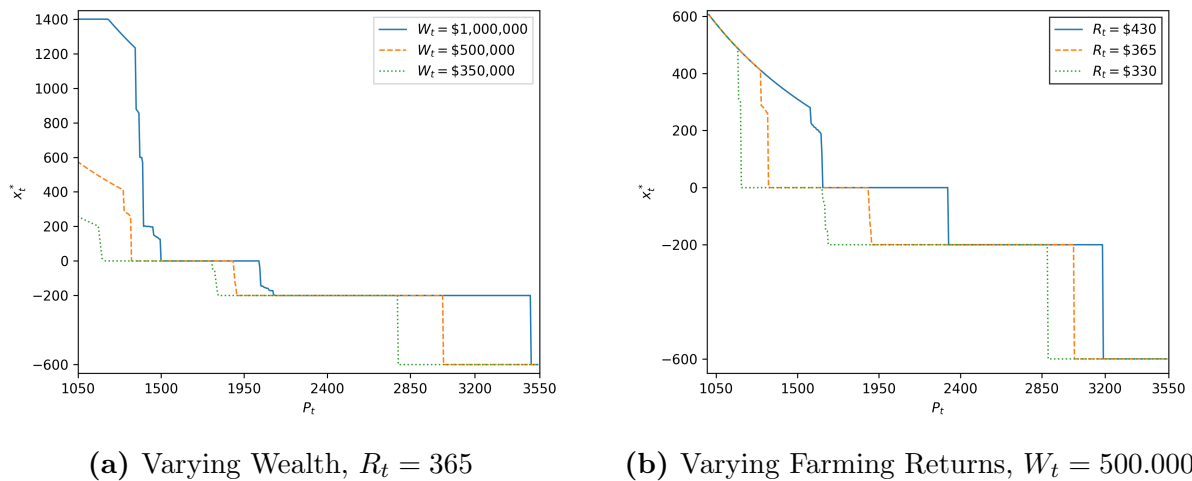
| Model               | Initial implementation | Vectorized + Numba | Speedup              |
|---------------------|------------------------|--------------------|----------------------|
| Without mutual fund | 9 min 22.4 sec         | 6.1 sec            | $\approx 92.2\times$ |
| With mutual fund    | 2 hr 36 min 48.7 sec   | 1 min 53.0 sec     | $\approx 83.3\times$ |

**Table 2:** Speed-up from initial implementation to final implementation

## 4 Replication Results

### 4.1 Model without mutual funds

In figure 1, we were able to reproduce the optimal farmland investment plots of Lohano and King (2009). The plots show the optimal farmland purchase at period  $t = 0$  as a function of farmland price, for different levels of farmer wealth and return per acre, for a farmer with an initial holding of 600 acres of farmland. In figure 1a, the return per acre is fixed to 365\$. This figure shows that if the farmer has 500,000 dollars and the price of land is cheap (1050\$), the optimal choice is to purchase around 600 acres of land. As farmland prices,  $P_t$ , increases ( $\sim 1300$ \$) the optimal policy transitions to stop investing in new farmland and after the price of an acre of land rises above 1900\$ the optimal choice is to start selling their farmland. If the price of farmland reaches above 3050 the farm manager will optimally sell all of his land. The upper- and lower-class farmers have similar patterns but the wealthy farmer will buy more land and be in the market for increasing land prices, while the lower class farm manager will stop investing and start selling off land earlier than the middle-class farmer.



**Figure 1:** Optimal farmland investment policy without mutual fund investment. Figure (a) shows the policy for different wealth levels with  $R_t = 365$ , while figure (b) shows the policy for different farming returns with  $W_t = 500,000$ .

When varying the return on farmland in figure 1b, with fixed wealth at 500,000, then a higher return results in the farmers to invest more in the farmland at higher farmland prices. The authors suggest, the ranges of inaction is mainly due to the transaction costs on buying and selling farm assets, as well as the financial constraint of a binding debt-to-farm-assets which prevents land purchase. The claim of the impact of transaction costs and financial constraint of debt-to-farm-ratio will be investigated in the counterfactuals.

| Year t | Initial DFA = 0.7        |                            |                          | Initial DFA = 0.57       |                           |                          | Initial DFA = 0.14       |                            |                          |
|--------|--------------------------|----------------------------|--------------------------|--------------------------|---------------------------|--------------------------|--------------------------|----------------------------|--------------------------|
|        | Expected net wealth (\$) | Expected farm size (acres) | Prob. of exiting farming | Expected net wealth (\$) | Expected farmsize (acres) | Prob. of exiting farming | Expected net wealth (\$) | Expected farm size (acres) | Prob. of exiting farming |
| 0      | 350,000                  | 600                        | 0.000                    | 500,000                  | 600                       | 0.000                    | 1,000,000                | 600                        | 0.000                    |
| 1      | 374,820                  | 600                        | 0.000                    | 538,427                  | 600                       | 0.000                    | 1,088,859                | 600                        | 0.000                    |
| 2      | 399,147                  | 511                        | 0.006                    | 583,060                  | 541                       | 0.000                    | 1,164,002                | 763                        | 0.000                    |
| 3      | 433,949                  | 485                        | 0.016                    | 620,764                  | 548                       | 0.000                    | 1,243,864                | 880                        | 0.000                    |
| 4      | 471,061                  | 484                        | 0.032                    | 661,380                  | 563                       | 0.000                    | 1,369,816                | 973                        | 0.000                    |
| 5      | 503,755                  | 498                        | 0.048                    | 720,353                  | 596                       | 0.000                    | 1,484,942                | 1,058                      | 0.000                    |
| 6      | 556,993                  | 529                        | 0.064                    | 775,173                  | 646                       | 0.000                    | 1,606,477                | 1,142                      | 0.000                    |
| 7      | 603,354                  | 543                        | 0.086                    | 845,891                  | 699                       | 0.006                    | 1,765,178                | 1,158                      | 0.000                    |
| 8      | 659,201                  | 581                        | 0.098                    | 939,163                  | 732                       | 0.008                    | 1,934,259                | 1,178                      | 0.000                    |
| 9      | 718,614                  | 613                        | 0.110                    | 1,036,676                | 781                       | 0.014                    | 2,067,609                | 1,188                      | 0.000                    |
| 10     | 764,629                  | 640                        | 0.144                    | 1,136,920                | 831                       | 0.018                    | 2,210,852                | 1,207                      | 0.000                    |

**Table 3:** Stochastic dynamic path. Initial parameter values:  $R_0 = 365$ ,  $P_0 = 1,775$ .

Table 3, is based on Monte Carlo simulations, by drawing 500 trials for each of the 20 years. In each trial, shocks were drawn for the error terms of the two stochastic variables: gross returns and farmland prices. Applying these shocks and following the optimal policy in each period generates 500 simulated paths, representing the distribution of states over time. The expected path was then obtained by averaging the simulated outcomes for each year. Our simulations largely coincide with Lohano and King (2009), with smaller differences due to randomness.

## 4.2 Model with mutual funds

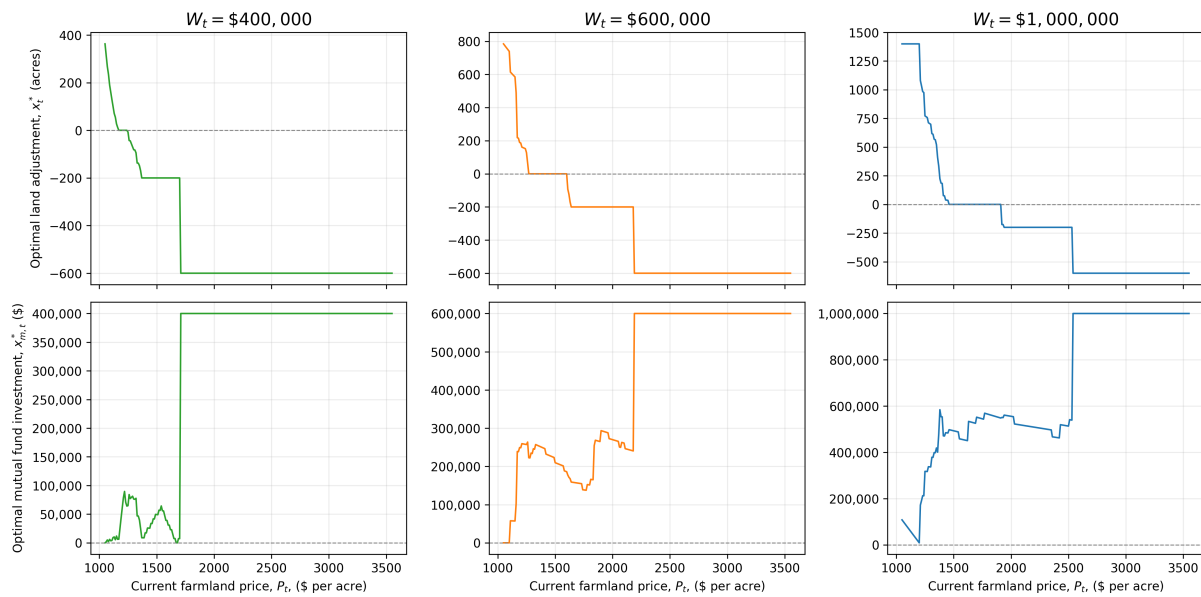
|                             |                 |                   | Without Mutual                       |                     | With Mutual                          |                                     |                     |
|-----------------------------|-----------------|-------------------|--------------------------------------|---------------------|--------------------------------------|-------------------------------------|---------------------|
| Current farmland price (\$) | Net wealth (\$) | Initial DFA ratio | Optimal farmland transaction (acres) | Resulting DFA ratio | Optimal farmland transaction (acres) | Optimal mutual-fund investment (\$) | Resulting DFA ratio |
| 1500                        | 400,000         | 0.61              | -200                                 | 0.41                | -200                                 | 49,330                              | 0.48                |
| 1500                        | 600,000         | 0.41              | 0                                    | 0.41                | 0                                    | 209,652                             | 0.61                |
| 1500                        | 1,000,000       | 0.01              | 0                                    | 0.01                | 0                                    | 492,986                             | 0.50                |
| 1775                        | 400,000         | 0.66              | -200                                 | 0.49                | -600                                 | 400,000                             | 0.00                |
| 1775                        | 600,000         | 0.49              | -200                                 | 0.23                | -200                                 | 137,362                             | 0.41                |
| 1775                        | 1,000,000       | 0.14              | 0                                    | 0.14                | 0                                    | 568,269                             | 0.63                |

**Table 4:** Optimal investment policy. Initial parameter values:  $R_0 = 365$ ,  $L_0 = 600$ .

Table 4 presents the replicated optimal investment policies for models with and without mutual fund access. Overall, the replication reproduces the main findings of the original study. In both models, optimal farmland investment decreases as farmland prices rise and increases with net wealth. As farmland becomes more expensive, larger land sales are observed, particularly at lower wealth levels, while higher net wealth reduces the need to liquidate farmland. Table 4 also confirms that access to mutual fund investment generally lowers direct farmland investment relative to the model without mutual funds, indicating

that financial assets partly substitute for farmland holdings. In the most extreme case, when farmland prices are \$1,775 and net wealth is \$400,000, the optimal decision in the mutual fund model is to sell all 600 acres, implying a complete exit from farming. This result suggests that when farmland prices are high and financial wealth is relatively low, reallocating assets entirely into financial investments becomes optimal.

Access to mutual funds also tends to increase the resulting debt-to-farm-asset ratio, consistent with the authors' conclusion that diversification opportunities make greater leverage attractive. The numerical values in Table 4 are exact to Lohano and King (2009).



**Figure 2:** Optimal policy with mutual fund. Initial parameter values:  $R_0 = 365$ ,  $L_0 = 600$ .

Figure 2 plots the optimal policy functions for farmland and mutual fund investment for different levels of initial wealth. The policy for farmland investment follows a broadly similar pattern to the model without mutual fund investment, with regions of inaction and disinvestment at high farmland prices. However, compared to the model without mutual funds, the inaction and selling regions occur at lower farmland prices. This suggests that the availability of a mutual fund introduces an alternative investment opportunity, reducing the relative attractiveness of farmland investment when farmland prices rise.

As farmland prices increase, the farmer increasingly reallocates wealth away from farmland and toward mutual fund investment. The figure also shows heterogeneity across wealth levels. Wealthier farmers allocate significantly more wealth to mutual funds and rely less heavily on debt financing, whereas lower-wealth farmers maintain larger leveraged farmland positions. This suggests that access to financial diversification increases with wealth, while borrowing constraints remain more relevant for poorer farmers.

|      | Without Mutual Fund      |                             |                           |                          | With Mutual Fund         |                             |                           |                          |
|------|--------------------------|-----------------------------|---------------------------|--------------------------|--------------------------|-----------------------------|---------------------------|--------------------------|
| Year | Expected net wealth (\$) | St. dev. of net wealth (\$) | Expected farmland (acres) | Prob. of exiting farming | Expected net wealth (\$) | St. dev. of net wealth (\$) | Expected farmland (acres) | Prob. of exiting farming |
| 0    | 1,000,000                | -                           | 600                       | 0.000                    | 1,000,000                | -                           | 600                       | -                        |
| 1    | 1,063,820                | 140,345                     | 600                       | 0.000                    | 1,069,813                | 168,529                     | 600                       | 0.000                    |
| 2    | 1,126,407                | 204,423                     | 616                       | 0.000                    | 1,144,909                | 268,361                     | 654                       | 0.002                    |
| 3    | 1,211,066                | 260,735                     | 673                       | 0.000                    | 1,241,602                | 353,393                     | 743                       | 0.004                    |
| 4    | 1,299,291                | 347,838                     | 745                       | 0.000                    | 1,359,770                | 465,528                     | 834                       | 0.020                    |
| 5    | 1,394,016                | 452,739                     | 816                       | 0.000                    | 1,492,199                | 589,715                     | 894                       | 0.052                    |
| 6    | 1,520,042                | 558,679                     | 880                       | 0.000                    | 1,624,063                | 720,736                     | 931                       | 0.064                    |
| 7    | 1,639,761                | 676,047                     | 958                       | 0.004                    | 1,777,388                | 841,106                     | 946                       | 0.090                    |
| 8    | 1,767,063                | 768,900                     | 1,045                     | 0.004                    | 1,965,490                | 988,901                     | 974                       | 0.102                    |
| 9    | 1,915,917                | 864,453                     | 1,077                     | 0.006                    | 2,152,876                | 1,142,193                   | 955                       | 0.116                    |
| 10   | 2,074,559                | 965,737                     | 1,121                     | 0.010                    | 2,368,074                | 1,286,069                   | 948                       | 0.132                    |

**Table 5:** Stochastic dynamic path with and without mutual fund investment. Initial parameter values:  $R_0 = \$365$ ,  $P_0 = \$1,775$ ,  $L_0 = 600$  and  $W_0 = \$1,000,000$ .

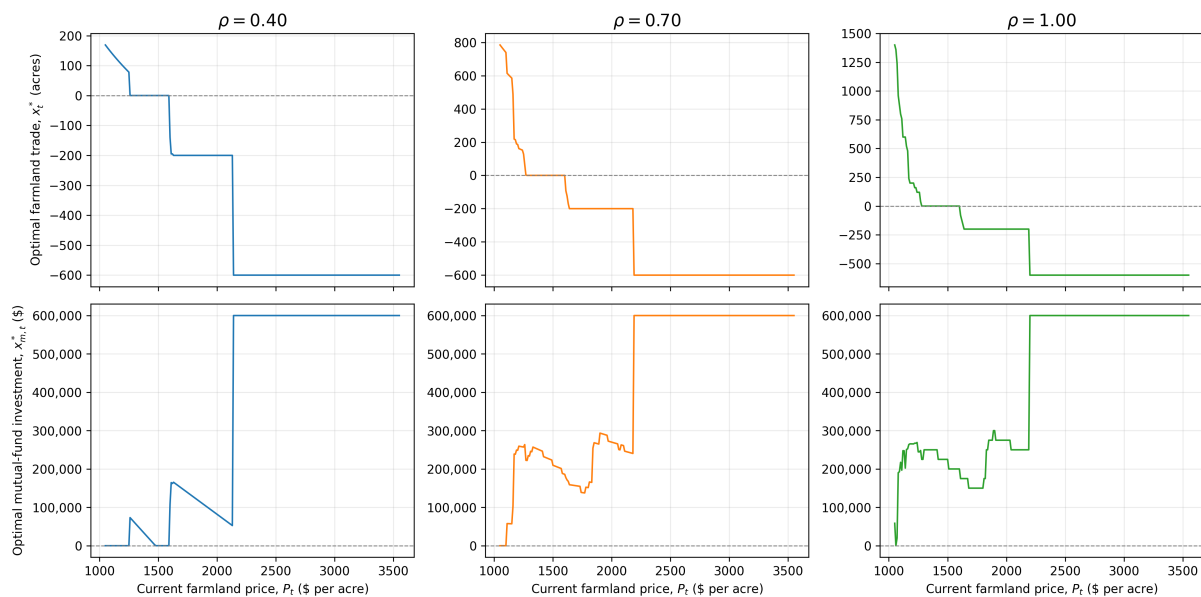
Table 5 presents the replicated expected path and probabilities with mutual fund investment. The table is produced by Monte Carlo simulations as introduced earlier. Because the simulations depend on Monte Carlo draws, an exact replication is not possible. However, the table closely matches Lohano and King (2009) findings, where the probability of exiting farming increases with the introduction of mutual fund investment. Furthermore, the expected farm size in year 10 differs in the two models as expected farm size is smaller when they have the option to invest in the mutual fund. Additionally, the path of expected net wealth is larger when mutual fund investment is introduced, however the standard deviation of net wealth is larger. Our mutual fund findings have slightly lower expected wealth, lower standard deviations and more expected farmland leading to a slightly lower probability of a farmer exiting the market than the findings of Lohano and King (2009), but the same conclusions remain.

## 5 Counterfactuals

To challenge and extend the original authors conclusions, we perform a number of different counterfactuals. Specifically, we investigate the optimal policy of inaction, which Lohano and King (2009) argues is caused by transactions costs and borrowing constraints. By inaction, we refer to regions of the policy function where the farmer optimally chooses not to adjust farmland holdings despite changes in farmland prices or wealth ( $x^* = 0$ ). We therefore investigate this matter and see what has an effect on the optimal policy of inaction. Hence, the following section will include a change in the borrowing constraint, eliminating transaction costs, and extending the model to include economies of scale.

### 5.1 Borrowing constraint

In section 4.1 we mentioned that Lohano and King (2009) argued the optimal policy of inaction was mainly due to transactions costs and the financial constraint imposed by a binding DFA ratio. In figure 2, we saw the same pattern for the model with mutual fund investment. Hence, we investigate the effect of the borrowing constraint by varying the maximum debt allowed through different values of  $\rho$ .



**Figure 3:** Optimal policy with mutual fund investment for varying values of  $\rho$ . Initial parameter values:  $R_0 = 365$ ,  $L_0 = 600$ ,  $W_0 = 600,000$ .

Figure 3 plots the optimal policy at  $t = 0$  for different values of  $\rho$ . Notably, the regions of inaction remain relatively stable across different borrowing limits, suggesting that the borrowing constraint is not the primary driver of farmland investment inaction. Instead, the debt limit primarily affects expansion decisions when farmland prices are low. In these cases, a higher value of  $\rho$  allows the farmer to borrow more and therefore invest more aggressively in farmland. By contrast, disinvestment behaviour at high farmland

prices remains relatively similar across values of  $\rho$ . As farmland prices increase, wealth is increasingly reallocated from farmland toward mutual fund investment.

| Year | Expected wealth (\$) |              |              | Std. dev. wealth (\$) |              |              | Expected farmland |              |              | Prob. exit   |              |              |
|------|----------------------|--------------|--------------|-----------------------|--------------|--------------|-------------------|--------------|--------------|--------------|--------------|--------------|
|      | $\rho = 0.4$         | $\rho = 0.7$ | $\rho = 1.0$ | $\rho = 0.4$          | $\rho = 0.7$ | $\rho = 1.0$ | $\rho = 0.4$      | $\rho = 0.7$ | $\rho = 1.0$ | $\rho = 0.4$ | $\rho = 0.7$ | $\rho = 1.0$ |
| 0    | 600,000              | 600,000      | 600,000      | —                     | —            | —            | 600               | 600          | 600          | 0.000        | 0.000        | 0.000        |
| 1    | 640,472              | 640,512      | 640,618      | 92,526                | 92,765       | 93,430       | 400               | 400          | 400          | 0.000        | 0.000        | 0.000        |
| 2    | 687,508              | 689,636      | 688,453      | 136,746               | 144,837      | 146,820      | 372               | 395          | 400          | 0.094        | 0.064        | 0.056        |
| 3    | 743,980              | 744,378      | 745,463      | 170,358               | 184,704      | 197,193      | 353               | 411          | 424          | 0.184        | 0.142        | 0.132        |
| 4    | 806,180              | 812,250      | 818,463      | 206,619               | 236,302      | 260,492      | 347               | 434          | 471          | 0.254        | 0.202        | 0.196        |
| 5    | 884,288              | 901,526      | 909,668      | 251,192               | 316,156      | 342,901      | 344               | 478          | 514          | 0.326        | 0.248        | 0.250        |
| 6    | 957,213              | 976,484      | 996,005      | 287,101               | 357,114      | 409,348      | 347               | 513          | 552          | 0.366        | 0.290        | 0.288        |
| 7    | 1,044,339            | 1,092,474    | 1,115,139    | 341,838               | 455,732      | 549,526      | 363               | 559          | 566          | 0.388        | 0.318        | 0.310        |
| 8    | 1,127,108            | 1,193,242    | 1,208,397    | 386,811               | 524,521      | 617,422      | 378               | 561          | 583          | 0.416        | 0.354        | 0.328        |
| 9    | 1,223,259            | 1,316,314    | 1,324,691    | 440,956               | 622,024      | 686,482      | 389               | 578          | 600          | 0.442        | 0.386        | 0.356        |
| 10   | 1,329,995            | 1,440,380    | 1,464,892    | 492,845               | 743,817      | 837,310      | 411               | 597          | 593          | 0.454        | 0.400        | 0.394        |

**Table 6:** Simulation results with varying values of  $\rho$

This behaviour is also reflected in the simulated paths shown in table 6. Relaxing the borrowing constraint through a higher value of  $\rho$  leads to higher expected wealth accumulation over the 10-year horizon, but also substantially increases the volatility of wealth outcomes. This illustrates a clear leverage-induced risk-return trade-off, where farmers with greater borrowing capacity are able to expand farmland holdings more aggressively and achieve higher expected returns, while simultaneously facing greater financial risk.

The simulations further show that higher borrowing capacity results in larger expected farmland holdings throughout the horizon, consistent with the policy functions in figure 3. Interestingly, despite the increase in wealth volatility, the probability of exiting farming declines slightly as  $\rho$  increases. This suggests that when the farmer can borrow more, the farmer can take advantage of good investment opportunities. However, higher leverage also makes wealth outcomes more uncertain.

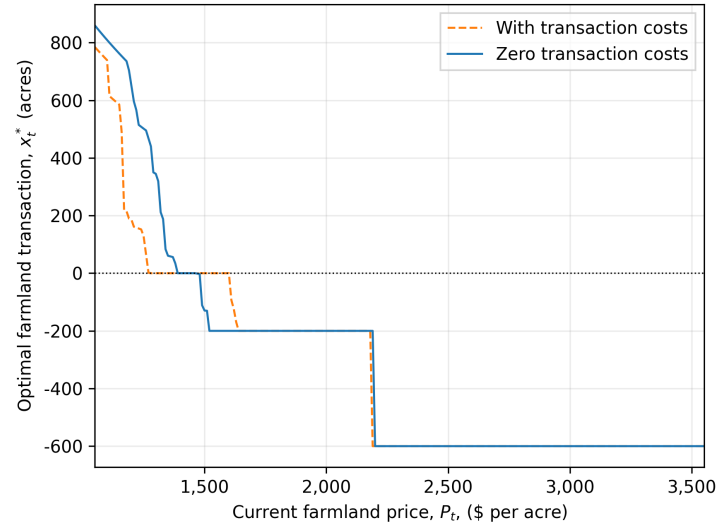
Overall, the results suggest that relaxing borrowing constraints primarily amplifies the scale and riskiness of investment decisions rather than fundamentally altering the structure of the optimal policy.

## 5.2 Zero transaction costs

In Section 5.1 we showed that changing  $\rho$  mainly affects land purchases at low prices, rather than the inactive zone. Therefore for this section, we keep  $\rho = 0.7$  fixed and focus on the effects from removing transaction costs.

We now look at what happens when the transaction costs are removed from the model. This is done by setting both buying and selling transaction costs equal to zero, thus  $tc_s = tc_p = 0$ . In the baseline model, it is costly for the farmer to buy or sell farmland, which can make it optimal to wait instead of adjusting land holdings. When we remove

these costs, we can investigate how the farmer adjusts his farmland holdings.



**Figure 4:** Optimal farmland policy with mutual fund investment for different farmland prices. Initial parameter values:  $R_t = 365$ ,  $L_t = 600$ ,  $W_t = 600,000$ , and  $\rho = 0.7$ . The figure compares the baseline model with transaction costs to the model with zero transaction costs.

Figure 4 shows that the farmer buys land when prices are low and sells land when prices are high. The baseline model with transaction costs has a larger inactive region, where  $x_t^* = 0$  compared to the model with zero transaction costs, where the inactive region is relatively small.

When transaction costs are removed, the farmer adjusts more actively. At low farmland prices, the farmer buys more land, and as prices increase, the farmer starts selling earlier. This is intuitive, because buying and selling land is cheaper when transaction costs are zero. Overall, the figure shows that transaction costs make the farmer less willing to adjust land holdings.

| Year | With Mutual Fund         |                             |                           |                          | Zero Transaction Costs   |                             |                           |                          |
|------|--------------------------|-----------------------------|---------------------------|--------------------------|--------------------------|-----------------------------|---------------------------|--------------------------|
|      | Expected net wealth (\$) | St. dev. of net wealth (\$) | Expected farmland (acres) | Prob. of exiting farming | Expected net wealth (\$) | St. dev. of net wealth (\$) | Expected farmland (acres) | Prob. of exiting farming |
| 0    | 1,000,000                | -                           | 600                       | -                        | 1,000,000                | -                           | 600                       | 0.000                    |
| 1    | 1,069,813                | 168,529                     | 600                       | 0.000                    | 1,064,268                | 124,294                     | 430                       | 0.000                    |
| 2    | 1,144,909                | 268,361                     | 654                       | 0.002                    | 1,156,066                | 254,255                     | 682                       | 0.004                    |
| 3    | 1,241,602                | 353,393                     | 743                       | 0.004                    | 1,268,466                | 378,235                     | 788                       | 0.012                    |
| 4    | 1,359,770                | 465,528                     | 834                       | 0.020                    | 1,418,952                | 504,759                     | 868                       | 0.028                    |
| 5    | 1,492,199                | 589,715                     | 894                       | 0.052                    | 1,579,070                | 620,381                     | 948                       | 0.048                    |
| 6    | 1,624,063                | 720,736                     | 931                       | 0.064                    | 1,737,578                | 758,296                     | 973                       | 0.060                    |
| 7    | 1,777,388                | 841,106                     | 946                       | 0.090                    | 1,878,170                | 862,737                     | 969                       | 0.078                    |
| 8    | 1,965,490                | 988,901                     | 974                       | 0.102                    | 2,061,569                | 983,255                     | 995                       | 0.100                    |
| 9    | 2,152,876                | 1,142,193                   | 955                       | 0.116                    | 2,252,152                | 1,073,721                   | 985                       | 0.120                    |
| 10   | 2,368,074                | 1,286,069                   | 948                       | 0.132                    | 2,472,936                | 1,253,353                   | 1,013                     | 0.138                    |

**Table 7:** Comparison of the baseline model with mutual funds and the zero transaction cost model with  $\rho = 0.7$ .

Table 7 shows how removing transaction costs affects the simulated path over time. In year 1, expected farmland falls from 600 acres in the baseline model to 430 acres in the zero transaction costs model. This suggests that the farmer makes a large immediate adjustment when buying and selling land is no longer costly.

Over time, expected farmland becomes higher when transaction costs are removed. In year 10, the expected net wealth increases from 2,368,074 in the baseline model to 2,472,936 in the zero transaction costs model. Expected farmland also increases from 948 acres to 1,013 acres, however the probability of exiting is almost unchanged and only increases from 0.132 to 0.138.

Overall, this counterfactual shows that removing transaction costs mainly affects how much farmland the farmer holds and not the probability of exit. Farmland holdings are higher in the long run, while exit risk is almost unchanged.

### 5.3 Economies of scale

Lohano and King (2009) assume that production costs scale linearly with the amount of farmland operated. In particular, the baseline specification assumes total operating costs equal to  $cL_{t+1}$ , implying a constant marginal and average cost per acre regardless of farm size. However, in practice, larger farms often benefit from economies of scale due to more efficient use of machinery, lower administrative costs per acre, improved logistics, and stronger bargaining power when purchasing inputs (Duffy 2009).

To capture these effects, we extend the model by introducing a concave production cost function. Specifically, we replace the linear cost term  $cL_{t+1}$  with:

$$C(L_{t+1}) = cL_{ref}^{1-\eta}L_{t+1}^{\eta}, \quad \eta \in (0, 1]$$

where  $\eta$  governs the degree of economies of scale. When  $\eta = 1$ , the specification collapses to the original linear cost function and there are no economies of scale. Lower values of  $\eta$  imply stronger economies of scale, since total costs increase less than proportionally with farm size. The term  $L_{ref}$  denotes a reference farm size and is included as a normalization constant. This normalization ensures that total production costs remain equal to  $cL_{ref}$  when  $L_{t+1} = L_{ref}$  regardless of the value of  $\eta$ .

This allows us to isolate the effect of economies of scale without simultaneously changing costs at the reference farm size. We set  $L_{ref} = 600$ , corresponding to the farmer's initial land holdings in the baseline simulations.

The modified specification implies decreasing average and marginal costs with respect to farm size. Average production costs per acre are given by:

$$\frac{C(L_{t+1})}{L_{t+1}} = cL_{ref}^{1-\eta}L_{t+1}^{\eta-1}$$

which decreases with farm size whenever  $\eta < 1$ . Similarly, marginal production costs are:

$$\frac{\partial C(L_{t+1})}{\partial L_{t+1}} = c\eta L_{ref}^{1-\eta}L_{t+1}^{\eta-1}$$

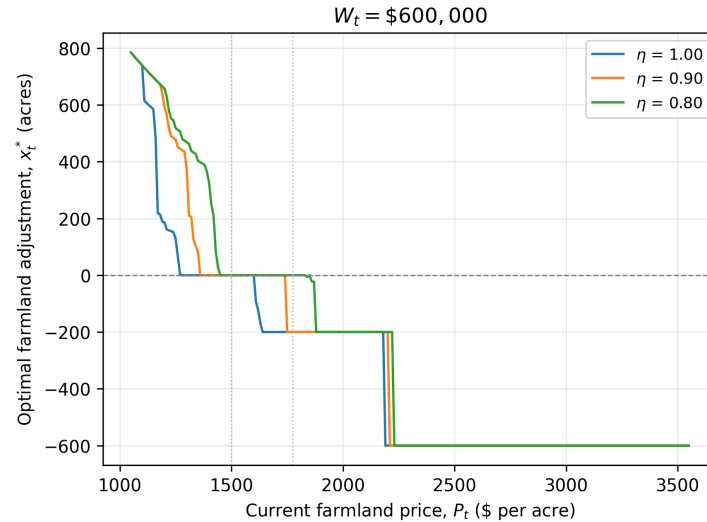
which also decline with farm size when  $\eta < 1$ . Economically, this implies that larger farms can operate additional acres at lower marginal cost than smaller farms.

Introducing economies of scale changes the dynamic incentives within the model. Since larger farms face relatively lower operating costs, they generate higher net returns and accumulate more wealth. This creates a self-reinforcing mechanism where successful farms can expand further over time, leading to greater land concentration and lower probabilities of exit. Economies of scale also interact with the borrowing constraint. Larger farms become better able to sustain leverage and satisfy the debt-to-farm-asset constraint, which further strengthens incentives to expand farmland holdings.

| $\eta$ | Year 10<br>expected Wealth | Year 10<br>std. dev of net Wealth | Year 10<br>expected farmland | Year 10<br>expected exit prob. | avg exit probability<br>(Y1-Y10) |
|--------|----------------------------|-----------------------------------|------------------------------|--------------------------------|----------------------------------|
| 1.0    | 1,440,380                  | 743,817                           | 597                          | 0.400                          | 0.240                            |
| 0.9    | 1,532,470                  | 977,957                           | 729                          | 0.384                          | 0.231                            |
| 0.8    | 1,607,443                  | 1,126,953                         | 855                          | 0.326                          | 0.194                            |

**Table 8:** Simulation results with economies of scale. Initial parameter values:  
 $L_0 = 600$ ,  $W_0 = 600,000$ ,  $P_0 = 1775$ ,  $R_0 = 365$

Table 8 reports simulation results for different values of  $\eta$ . The simulations are conducted using the baseline calibration with  $\rho = 0.7$ , default transaction costs, initial wealth of 600,000 \$ and initial land holdings of 600 acres in the model including mutual-funds. As economies of scale become stronger (lower  $\eta$ ), farmers optimally choose to expand farmland holdings substantially. Reducing  $\eta$  from 1.0 to 0.8 increases expected farmland holdings in year 10 from 597 to 855 acres, corresponding to an increase of approximately 48%. Expected wealth also rises from \$1.44 million to \$1.61 million. At the same time, the probability of exiting farming declines from 40% to 32.6%. The average exit probability across years 1-10 falls from 24.0% to 19.4%.



**Figure 5:** Optimal farmland purchase policy under different degrees of economies of scale. Initial parameter values:  $L_0 = 600$ ,  $W_0 = 600,000$ ,  $R_0 = 365$ .

The increase in the standard deviation of net wealth further suggests that economies of scale amplify heterogeneity across farmers. Since larger farms benefit disproportionately from lower average operating costs, successful farms accumulate wealth more rapidly and expand further over time, leading to greater dispersion in wealth outcomes and larger farmland holdings among successful farms.

Figure 5 illustrates how economies of scale affect optimal land adjustment decisions across different land prices. Lower values of  $\eta$  shift the policy function upward, indicating that farmers choose to purchase and retain larger amounts of farmland even when land prices increase. This reflects the fact that larger farms face lower average operating costs under stronger economies of scale, which raises the profitability of expansion.

Overall, the counterfactual suggests that even moderate economies of scale can substantially affect farmland concentration, wealth accumulation, and farm survival. The results are consistent with the theory behind economies of scale, where larger productions often benefit from lower average operating costs and therefore tend to expand over time.

## 6 Discussion and Conclusion

This paper aimed to replicate the findings of Lohano and King (2009) and extend the analysis of optimal farmland investment and portfolio choice for farmers in Minnesota. Using a Python implementation of the original stochastic dynamic programming model, we replicated the main policy functions and simulation results of the original paper and used the model to investigate how borrowing constraints, transaction costs, and economies of scale affect farmland investment decisions, wealth accumulation, and the probability of exiting farming.

Overall, the replication was successful and reproduced the main qualitative findings of Lohano and King (2009). In both the model with and without mutual fund access, optimal farmland investment decreases as farmland prices increase and increases with farmer wealth and expected farming returns. The replication also confirmed that access to mutual funds partly substitutes for farmland investment, as farmers allocate a larger share of wealth toward financial assets when diversification opportunities are available. Furthermore, the simulations showed that introducing mutual funds increases expected wealth accumulation, but also increases wealth volatility and the probability of exiting farming.

The counterfactual analysis provided several additional insights. Relaxing borrowing constraints mainly affected the scale and riskiness of investment decisions, allowing farmers to expand farmland holdings more aggressively while increasing wealth volatility through greater leverage exposure. By contrast, removing transaction costs substantially reduced the inaction region in the farmland adjustment policy, suggesting that transaction costs are an important source of persistence in farmland investment decisions.

Introducing economies of scale generated changes in both investment behaviour and long-run outcomes. Lower average operating costs for larger farms increased incentives to expand farmland holdings, leading to higher expected wealth accumulation and lower probabilities of exiting farming. At the same time, the increase in the standard deviation of wealth suggests that economies of scale amplify heterogeneity across farmers and contribute to greater concentration in farmland holdings over time.

Despite the successful replication and extensions, the model remains subject to several limitations. The framework assumes exogenous farmland prices and farming returns, and therefore does not capture general equilibrium effects or endogenous price formation in farmland markets. In addition, the model abstracts from several important features of agricultural production, including labour decisions, heterogeneous farmer productivity and climate risk. The finite planning horizon and irreversible exit assumption may also affect the long-run dynamics of farmland investment decisions.

Overall, the results highlight the importance of financial constraints, transaction costs, and economies of scale in shaping farmland investment behaviour. Transaction costs generate persistence in farmland holdings, borrowing constraints affect the scale and riskiness of investment decisions, and economies of scale create incentives for farm expansion and wealth concentration. More broadly, the results illustrate how dynamic programming methods can be used to study how financial frictions and scale economies shape long-run structural change in agriculture.

## References

- Asheim, Geir B and Kjell Arne Brekke (1993). *Sustainability when resource management has stochastic consequences*. Tech. rep. Discussion Papers.
- Duffy, Michael (2009). “Economies of size in production agriculture”. In: *Journal of hunger & environmental nutrition* 4.3-4, pp. 375–392.
- Lohano, Heman D. and Robert P. King (2009). “A Stochastic Dynamic Programming Analysis of Farmland Investment and Financial Management”. In: *Canadian Journal of Agricultural Economics/Revue canadienne d’agroeconomie*.
- Taff, Steven J (2008). “Minnesota Farm Real Estate Sales: 1990-2007”. In.

## A Appendix

### A.1 Variable and parameter explanations

| Variable/Parameter | Definition                                                                   |
|--------------------|------------------------------------------------------------------------------|
| $R_t$              | Return from farming per acre in period $t$                                   |
| $P_t$              | Price of farmland per acre in period $t$                                     |
| $M_t$              | Return on the risky financial asset in period $t$                            |
| $L_t$              | Land holdings (acres owned) in period $t$                                    |
| $W_t$              | Farmer's wealth in period $t$                                                |
| $A_t$              | Cash balance in period $t$                                                   |
| $x_t$              | Change in land holdings (purchases or sales) in period $t$                   |
| $x_{mt}$           | Investment in the risky financial asset                                      |
| $\kappa$           | Value of machinery and equipment per acre of land                            |
| $tc_p$             | Transaction cost per acre when purchasing farmland, machinery, and equipment |
| $tc_s$             | Transaction cost per acre when selling farmland, machinery, and equipment    |
| $r_b$              | Interest rate on borrowing (debt financing on farmland)                      |
| $r_l$              | Interest rate on lending (risk-free investment)                              |
| $c$                | Production cost per acre                                                     |
| $X_t$              | Set of all levels of control variable $x_t$                                  |
| $X_{mt}$           | Set of all levels of control variable $x_{mt}$                               |

**Table 9:** Definition of variables and parameters (reproduced from Lohano and King, 2009)

### A.2 Estimation of gross return equation

The gross return process is modeled as an autoregressive time series using the Box–Jenkins methodology. The selected specification is a first-order autoregressive model,  $AR(1)$ , estimated in logarithms by OLS, with diagnostics (Ljung–Box and Jarque–Bera tests) supporting the assumptions of white noise and normally distributed errors. Under these conditions, the OLS estimator is consistent and coincides with the conditional maximum likelihood estimator.

| Variable                                                                                     | Coefficient estimate | Standard error | $t$ -statistic |
|----------------------------------------------------------------------------------------------|----------------------|----------------|----------------|
| Constant                                                                                     | 1.511810**           | 0.656765       | 2.301904       |
| $\ln R_{t-1}$                                                                                | 0.742391**           | 0.112674       | 6.588854       |
| $\widehat{\text{Var}}[\varepsilon_{1t}] = 0.030186$ , $R^2 = 0.533$ , adjusted $R^2 = 0.521$ |                      |                |                |

\*\*Significant at 5% level on a two-tailed test with the critical value of 2.021.

**Table 10:** Estimation of gross return equation: Dependent variable  $\ln R_t$

### A.3 Estimation of price equation

The farmland price process follows Burt (1986) and is specified as an autoregressive model in logarithms with two lags and lagged returns. The equation is estimated by OLS, and diagnostic tests (Ljung–Box and Jarque–Bera) indicate a stable specification with white noise, normally distributed errors. Using the Burt and Taylor (1989) reduction, the model is rewritten as a first-order process in  $\ln P_{t-1}$  and  $\ln R_{t-1}$ , with appropriately adjusted coefficients and error variance.

| Regression results                                                                           |                      |                |                | Reduction of lagged variable                        |                      |
|----------------------------------------------------------------------------------------------|----------------------|----------------|----------------|-----------------------------------------------------|----------------------|
| Variable                                                                                     | Coefficient estimate | Standard error | $t$ -statistic | Variable                                            | Coefficient estimate |
| Constant                                                                                     | 0.3540               | 0.436          | 0.813          | Constant                                            | 0.215001             |
| $\ln P_{t-1}$                                                                                | 1.4955**             | 0.127          | 11.787         | $\ln P_{t-1}$                                       | 0.908361             |
| $\ln P_{t-2}$                                                                                | -0.6463**            | 0.123          | -5.255         |                                                     |                      |
| $\ln R_{t-1}$                                                                                | 0.1308*              | 0.073          | 1.794          | $\ln R_{t-1}$                                       | 0.079432             |
| $\widehat{\text{Var}}[\varepsilon_{1t}] = 0.010068$ , $R^2 = 0.929$ , adjusted $R^2 = 0.922$ |                      |                |                | $\widehat{\text{Var}}[\varepsilon_{2t}] = 0.017292$ |                      |

\*\*Significant at 5% level on a two-tailed test with the critical value of 2.021.

\*Significant at 5% level on a one-tailed test with the critical value of 1.684.

**Table 11:** Estimation of price equation: Dependent variable  $\ln P_t$

### A.4 Estimation of mutual fund equation

The mutual fund return equation is specified using the Box–Jenkins approach and estimated by OLS, with results reported in Table 4. Diagnostic tests (Ljung–Box and Jarque–Bera) support a stable specification with white noise and normally distributed errors. The results indicate that mutual fund returns are stochastic but not dependent on their lagged values, and therefore do not need to be included as a state variable in the DP model.

| Variable                                 | Coefficient estimate | Standard error | $t$ -statistic |
|------------------------------------------|----------------------|----------------|----------------|
| Constant                                 | 0.057800**           | 0.025654       | 2.253000       |
| $R^2 = -0.000$ , adjusted $R^2 = -0.000$ |                      |                |                |

\*\*Significant at 5% level on a two-tailed test with the critical value of 2.021.

**Table 12:** Estimation of mutual fund equation: Dependent variable  $\ln M_t$